

Probabilities on Graphs

Notes for Probabilistic Models of the Visual Cortex

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What this lecture covers

Graphs and weak coupling

Priors and double counting

Strong coupling

Explaining away

Model selection

A network for diagnosis

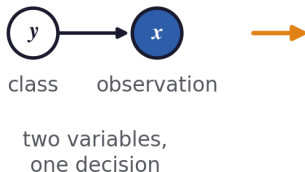
Summary

From Bayes decision theory to probabilities on graphs

Bayes decision theory was set out with two variables in view: an observation x and the quantity y that has to be decided. Vision rarely has only two. An image is produced by many scene variables at once, and several measurements are taken from it.

The new question is: how do several variables that influence each other combine into one distribution? The answer is a graph. The nodes specify the variables and the edges indicate which influence which.

Bayes decision theory



Probabilities on graphs

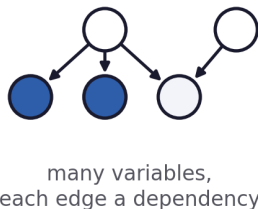


Figure 1: Two variables give one decision from one number. Many variables need a structure that says which of them act on which.

The general directed model - edges are directed

A directed graph has one node per variable and a directed edge from each variable to the variables it directly influences. Write $\text{pa}(\mu)$ for the parents of node μ , the nodes with an edge into it, and $\text{nd}(\mu)$ for its non-descendants, the nodes not reachable from μ by following arrows forwards. The model asserts that a node depends on its non-descendants only through its parents,

$$P(X_\mu \mid \vec{X}_{\text{pa}(\mu)}, \vec{X}_{\text{nd}(\mu)}) = P(X_\mu \mid \vec{X}_{\text{pa}(\mu)}),$$

and the joint distribution is then the product of one conditional distribution per node,

$$P(\{X_\mu : \mu \in \mathcal{V}\}) = \prod_{\mu \in \mathcal{V}} P(X_\mu \mid \vec{X}_{\text{pa}(\mu)}).$$

Descendants have to be left out. In $S \rightarrow C$ the node S has no parents, yet conditioning on its child C certainly changes what we believe about S : that is the next section. Arrows may not close a cycle, which would make a variable its own ancestor. A chain of states in time gives the Bayes-Kalman filter and the hidden Markov model.

The general directed model - edges are directed

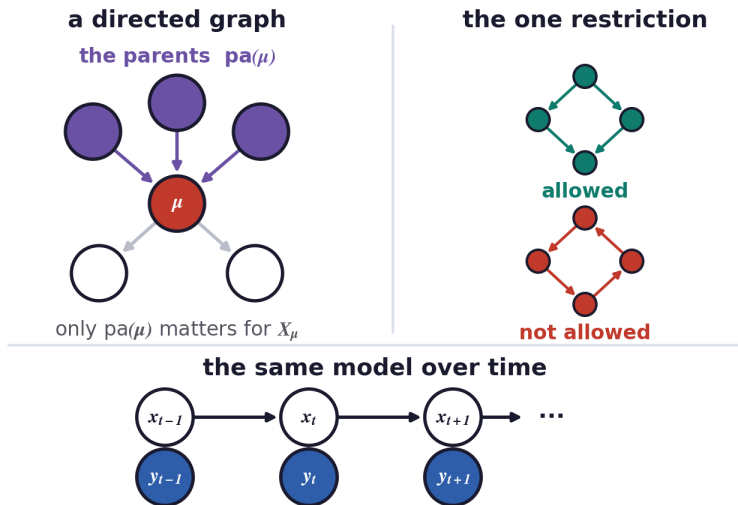


Figure 2: A node depends on the graph only through its parents. The same form covers a state evolving in time with one observation per step. Arrows that close a cycle are excluded.

One cue: a two-node graph

The smallest interesting graph has two nodes: a scene property S and a cue C measured from the image, with one edge $S \rightarrow C$. The edge carries the likelihood $P(C | S)$, which is the forward process: given the scene, what the measurement is likely to be. The node S carries the prior $P(S)$.

Estimating S from C runs against the edge, and the prior enters through Bayes' rule,

$$P(S | C) = \frac{P(C | S) P(S)}{P(C)}, \quad S^* = \arg \max_S P(C | S) P(S).$$

Slant from texture is one instance: S is the slant of the surface and C is the measured gradient of texture element size across the image.

One cue: a two-node graph

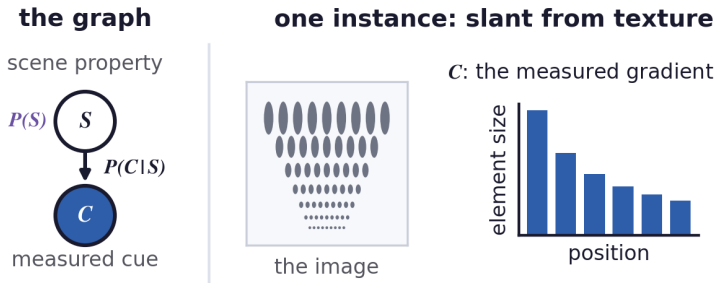


Figure 3: The edge runs from the scene to the measurement, and the estimate runs the other way. A slanted surface makes its texture elements shrink with distance, and that gradient is the cue.

Two cues: conditional independence

Add a second cue. Both are caused by the same scene property, so the graph is $C_1 \leftarrow S \rightarrow C_2$, with no edge between the two cues. That missing edge is the modelling assumption, and it says

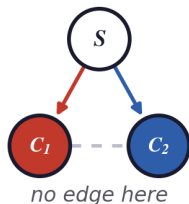
$$P(C_1, C_2 | S) = P(C_1 | S) P(C_2 | S).$$

Given S , knowing C_1 says nothing more about C_2 : the two cues carry independent evidence. Each cue then contributes its own likelihood factor, and the two factors multiply.

For a slanted textured surface seen with two eyes, C_1 is the texture gradient and C_2 is the binocular disparity. Both are produced by the slant, and the assumption is that the noise in one measurement is unrelated to the noise in the other.

Two cues: conditional independence

one common cause



The missing edge is the modelling assumption:

$$P(C_1, C_2 | S) = P(C_1 | S) P(C_2 | S)$$

Given S , knowing C_1 says nothing more about C_2 .

The two cues carry independent evidence about S .

An example: S is the surface slant, C_1 the texture gradient and C_2 the binocular disparity.

Figure 4: Weak coupling: the two cues are conditionally independent given S , and the prior on S is uniform.

Conditions for the linear rule

The linear rule takes the estimate to be a weighted sum of the two single-cue estimates,

$$S^* = \omega_1 S_1^* + \omega_2 S_2^*, \quad \omega_1 + \omega_2 = 1, \quad \omega_1, \omega_2 > 0.$$

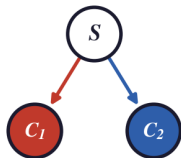
Three conditions together are the special case in which that rule is the optimal one.

- ▶ The cues are conditionally independent given S .
- ▶ Both likelihoods are Gaussian in S ,
 $P(C_i | S) \propto \exp\{-|C_i - S|^2/2\sigma_i^2\}$.
- ▶ The prior $P(S)$ is uniform, so it leaves every value of S equally supported and never moves the estimate.

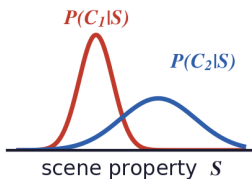
The three are sufficient, not necessary: correlated Gaussian cues under a Gaussian prior still give an estimate linear in the measurements, but with weights that are no longer the inverse variances. What is lost when one of the conditions is dropped is the simple average above. Sections 2 and 3 take away the third condition and then the first.

Conditions for the linear rule

1. cues independent given S



2. Gaussian likelihoods



3. uniform prior

flat: it never moves the estimate



Figure 5: The three conditions under which the optimal estimate is a weighted average of the two cues.

The optimal estimate from both cues

Under the three conditions the posterior is proportional to the product of the two likelihoods. Each cue on its own is best explained by its own measurement,

$$S_1^* = \arg \max_S P(C_1 | S) = C_1, \quad S_2^* = \arg \max_S P(C_2 | S) = C_2,$$

and the estimate from both is

$$S^* = \arg \max_S P(C_1 | S)P(C_2 | S) = \frac{\sigma_2^2}{\sigma_1^2 + \sigma_2^2} C_1 + \frac{\sigma_1^2}{\sigma_1^2 + \sigma_2^2} C_2.$$

The product of two Gaussians is a Gaussian whose mean is a weighted average of theirs, so the joint estimate lands between the two single-cue answers, nearer the narrower likelihood.

The optimal estimate from both cues

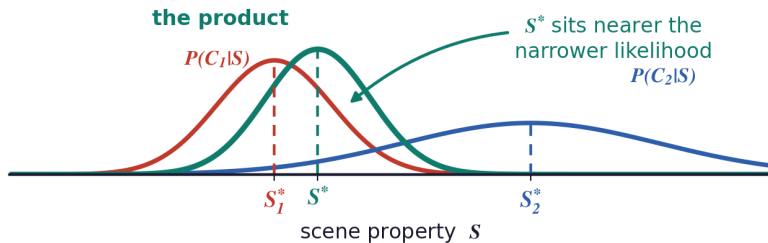


Figure 6: The product of the two likelihoods peaks between them, and closer to the one with the smaller variance.

Completing the square for two Gaussian likelihoods

Write $t_i = 1/\sigma_i^2$ for the precision of cue i . The exponent of the product is

$$-\frac{1}{2}t_1(S - C_1)^2 - \frac{1}{2}t_2(S - C_2)^2 = -\frac{1}{2}(t_1 + t_2)S^2 + (t_1C_1 + t_2C_2)S + \text{const.}$$

A quadratic $-\frac{1}{2}aS^2 + bS$ is $-\frac{a}{2}(S - b/a)^2$ up to a constant, so the product is Gaussian with

$$S^* = \frac{t_1C_1 + t_2C_2}{t_1 + t_2}, \quad \frac{1}{\sigma^{*2}} = t_1 + t_2.$$

Substituting $t_i = 1/\sigma_i^2$ turns the first expression into the weighted average $\omega_1C_1 + \omega_2C_2$ with $\omega_1 = \sigma_2^2/(\sigma_1^2 + \sigma_2^2)$. The second says the precisions add: the two cues together are more precise than either one, whatever the measurements turn out to be.

The weights are the inverse variances

Reading off the weights,

$$\omega_1 = \frac{\sigma_2^2}{\sigma_1^2 + \sigma_2^2} = \frac{t_1}{t_1 + t_2}, \quad \omega_2 = \frac{\sigma_1^2}{\sigma_1^2 + \sigma_2^2} = \frac{t_2}{t_1 + t_2}.$$

The cue with the smaller variance gets the larger weight, and a cue whose variance grows without limit is switched off. The combined variance $\sigma^{*2} = (t_1 + t_2)^{-1}$ is smaller than either σ_1^2 or σ_2^2 , so adding a cue can only sharpen the estimate.

This is the result that makes the model testable. It predicts the weights from the variances alone, with nothing left to fit: measure how noisy each cue is on its own, and the weight each cue should carry follows.

The weights are the inverse variances

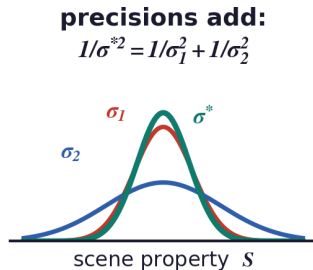
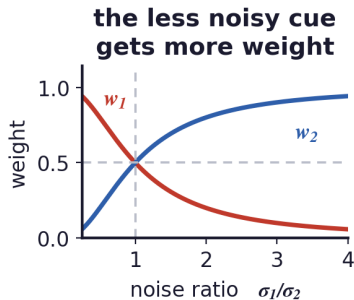


Figure 7: The weight of a cue falls as its noise grows, and the precision of the combination is the sum of the two precisions.

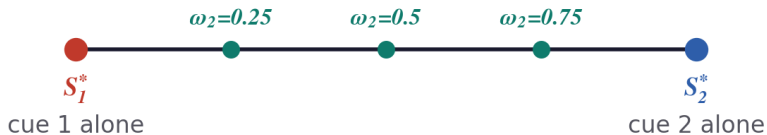
Landy et al.: the general weighted rule - an empirical finding

Any positive weights that sum to one define a combination rule, and the set of such rules is the segment between the two single-cue estimates. The inverse-variance weights are one point on that segment; a psychophysical experiment can measure which point a person uses.

Landy et al. reviewed the early cue-combination studies and argued that this form accounts for many of them. They also described gating, where one cue switches another off entirely, which is the end of the segment rather than an interior point. The inverse-variance rule approaches the same endpoint as one σ grows, but only in the limit: gating in their sense is a switch that is thrown, and the two are not the same mechanism.

any pair of weights is a combination rule

$S^* = \omega_1 S_1^* + \omega_2 S_2^*$ is a point on the segment



gating turns one cue off



Figure 8: Every pair of positive weights summing to one is a point on the segment between the two single-cue estimates, and gating is its endpoint.

Ernst and Banks: vision and touch

Subjects judged the height of a ridge seen in a stereo display and felt through force feedback. Noise added to the display raises the visual variance, and the visual weight falls with it. The measured weights follow the inverse-variance prediction.

The design is what makes the result strong. The two variances are measured first, in single-cue trials, and they fix the predicted weights before any two-cue trial is run. The two-cue trials then measure the weights directly, by putting a small conflict between vision and touch and seeing where the percept lands. The prediction has no free parameter.

Ernst and Banks: vision and touch

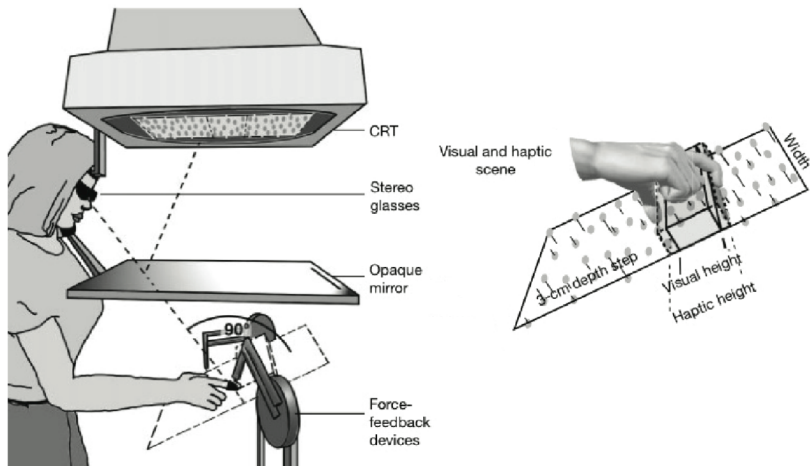


Figure 9: The apparatus. A stereo display and a force-feedback device present a visual and a haptic version of the same ridge.

The posterior for one cue

Take away the uniform prior. Each cue is modelled by its own likelihood, and both models use the same prior on S , because the prior describes the world and not the measurement. The two posteriors are

$$P(S | C_1) = \frac{P(C_1 | S) P(S)}{P(C_1)}, \quad P(S | C_2) = \frac{P(C_2 | S) P(S)}{P(C_2)},$$

and the two single-cue estimates are their maximisers,

$$S_1^* = \arg \max_S P(S | C_1), \quad S_2^* = \arg \max_S P(S | C_2).$$

Each of these estimates already contains the prior once. Taking the two and averaging them would therefore use the prior twice.

The posterior for one cue

the prior appears in each model



Figure 10: Two separate models, each carrying its own copy of the prior.

The posterior for both cues

The correct object is the posterior given both cues,

$$P(S | C_1, C_2) = \frac{P(C_1, C_2 | S) P(S)}{P(C_1, C_2)}.$$

Conditional independence turns the joint likelihood into one factor for each cue, which leaves

$$P(S | C_1, C_2) = \frac{\underbrace{P(C_1 | S) P(C_2 | S)}_{\text{one factor for each cue}} \underbrace{P(S)}_{\text{the prior, once}}}{P(C_1, C_2)}.$$

The prior multiplies in once, however many cues there are, because there is one scene and the prior is a statement about it.

The posterior for both cues

the prior appears once

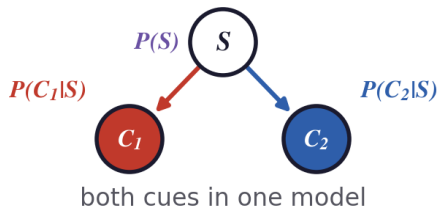


Figure 11: One model carrying both cues, and one copy of the prior.

A Gaussian prior enters only once

Make the prior Gaussian, centred on a value S_p with variance σ_p^2 , and write $t_p = 1/\sigma_p^2$ alongside the two cue precisions,

$$P(S) = \frac{1}{Z_p} \exp \left\{ -\frac{|S - S_p|^2}{2\sigma_p^2} \right\}, \quad t_1 = \frac{1}{\sigma_1^2}, \quad t_2 = \frac{1}{\sigma_2^2}, \quad t_p = \frac{1}{\sigma_p^2}.$$

Completing the square over three Gaussian factors instead of two gives the joint estimate, and over two factors gives each single-cue estimate,

$$S^* = \frac{t_1 C_1 + t_2 C_2 + t_p S_p}{t_1 + t_2 + t_p}, \quad S_1^* = \frac{t_1 C_1 + t_p S_p}{t_1 + t_p}, \quad S_2^* = \frac{t_2 C_2 + t_p S_p}{t_2 + t_p}.$$

Subscript p marks the prior. In this simplified scalar example S is one shape parameter and the prior mean S_p is its flattest value; each t is one over a variance, so a larger t means a more reliable source.

A Gaussian prior enters only once

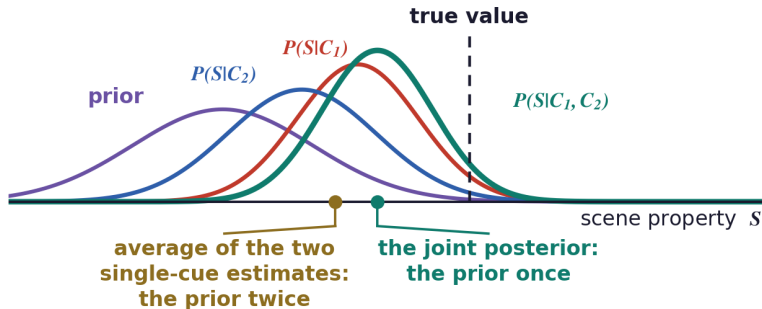


Figure 12: Each single-cue posterior is already pulled towards the prior. Their weighted average is pulled twice, and lands further from the truth than the joint posterior.

Why averaging the two estimates double counts

Take any weights $\omega_1 + \omega_2 = 1$ and average the two single-cue estimates. The prior mean appears in both, with total weight

$$\omega_1 \frac{t_p}{t_1 + t_p} + \omega_2 \frac{t_p}{t_2 + t_p},$$

while the joint estimate gives it weight $t_p/(t_1 + t_2 + t_p)$. The first is larger for every choice of weights, because each of its two terms already has a smaller denominator than the second.

The same reading works the other way round. The joint estimate has total precision $t_1 + t_2 + t_p$: two cues and one prior. An average of the two single-cue estimates behaves like two cues and two priors. So no pair of weights recovers the joint answer, and the gap grows as the prior gets tighter. When the prior is flat, $t_p = 0$, and the three expressions collapse to the weighted average of Section 1.

The prior biases each single-cue estimate

Shape from shading and shape from texture both assume the surface is smooth. A smoothness prior pulls each single-cue estimate of shape towards a flatter surface, so each of them understates the true slant. Any weighted average of two flattened estimates lies between them, and is therefore still flat.

This makes the double counting testable. If observers combined the two single-cue percepts linearly, the two-cue percept would have to lie between the two single-cue percepts. If instead they estimate from the joint posterior, the two-cue percept can lie outside that interval, closer to the truth than either cue alone.

The prior biases each single-cue estimate

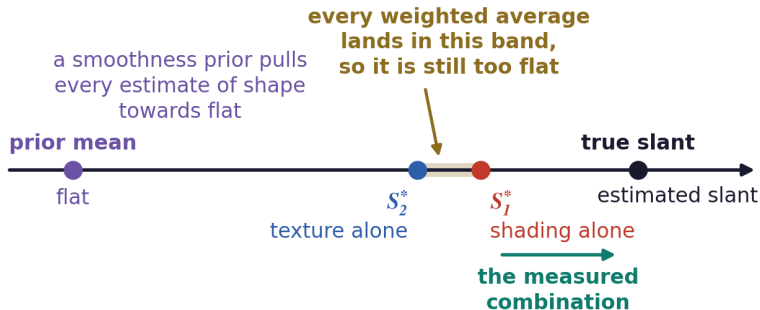


Figure 13: Both single-cue estimates are biased towards flat, so every weighted average of them is biased towards flat as well. The measured combination lies outside the band.

Shading and texture: the experiment

Subjects judged the elongation of an ellipsoid from shading only, from texture only, or from both. Each cue alone was seen as too flat, and the two together were close to correct.

That outcome rules out the linear rule for this pair of cues, because a weighted average of two flattened estimates cannot escape the band between them, and it is what the joint posterior predicts: the two likelihoods multiply and the single prior is applied once, so the combined estimate can exceed both single-cue estimates.

Shading and texture: the experiment

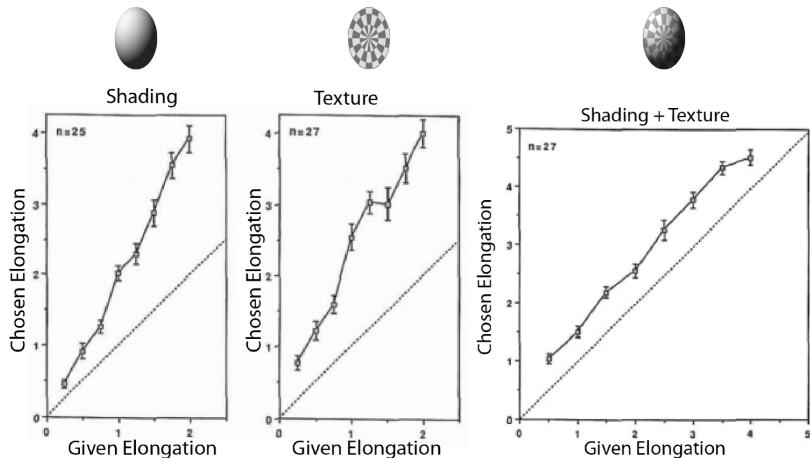


Figure 14: Chosen elongation against given elongation, for shading alone, texture alone, and the two together. The diagonal is a correct judgement. After Bülthoff and Mallot, 1990.

Weak and strong coupling

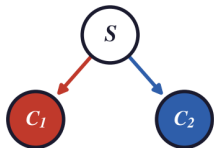
Weak coupling means the cues are conditionally independent given the scene, so the joint likelihood factorises into one term per cue. Add Gaussian likelihoods and a uniform prior and the linear rule of Section 1 is optimal; that is a special case of weak coupling, not its definition.

Strong coupling means the cues are dependent given the scene, because some further scene variable L feeds both of them. The likelihood no longer factorises and the estimate has to come from the joint model.

Two other things break the linear rule without breaking weak coupling: a shared non-uniform prior, which is Section 2, and a model that is itself in question, which is Section 5. The terminology is from Clark and Yuille, who set out the distinction and the conditions above.

Weak and strong coupling

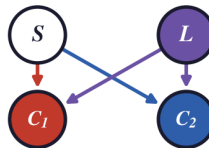
weak coupling



the cues are
independent given S

a linear rule is optimal

strong coupling



L makes the two
cues dependent

estimate S from the joint model

Figure 15: With a second scene variable L feeding both cues, the two cues are dependent given S on its own. There is still no edge between them; they are independent again once L is known too.

A taxonomy of cue interactions

Two patterns in the graph cover most of the cases.

- ▶ **Common cause.** One scene event produces two cues, and the evidence from them combines. This is the shape of Section 1.
- ▶ **Common effect.** Several scene variables produce one image measurement, and they have to be separated. This is the shape of Section 4.

The pattern in the graph decides how the cues must be treated, and it is visible before any number is written down. Relative depth producing both a shadow displacement and a stereo disparity is a common cause; an image measurement produced jointly by the object and the viewpoint is a common effect.

A taxonomy of cue interactions

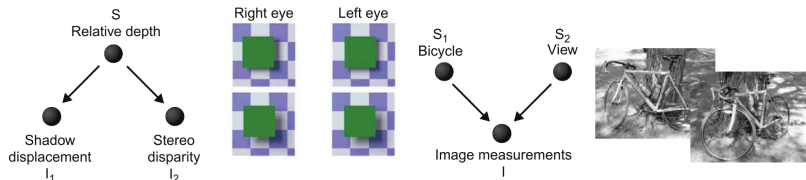


Figure 16: One scene variable producing two measurements, and two scene variables producing one measurement.

Visual cues depend on one another

The flying carpet needs the ground plane, the segmentation of the figure from the ground, the shadow and the lighting at once. These cues are computed from the same image regions, so they constrain one another: the ground plane fixes where a shadow may fall, and the lighting fixes what shape it may have.

Splitting a natural image into a texture part and a shading part is itself an open problem, which is the same point at the level of the measurement. The two cues of Section 1 were treated as separate numbers. In a real image they are two readings of the same pixels, and the noise in one is not independent of the noise in the other.

Visual cues depend on one another

the flying carpet the causal structure behind the percept

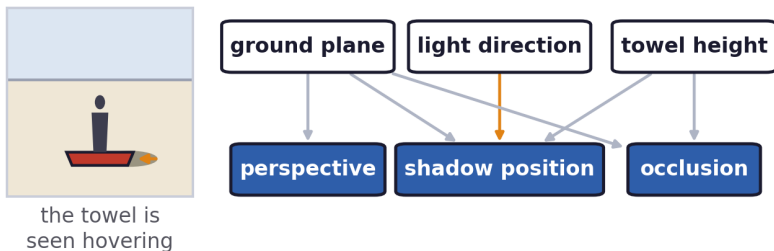


Figure 17: The same scene variables feed several cues, so the cues are dependent.

A cast shadow decides the perceived motion

The image trajectory of the ball is identical in the two movies, and only the shadow differs. In one the ball is seen sliding along the floor to the back of the box; in the other it is seen rising off the floor and staying at the same depth.

The visual system finds the floor, the ball and the light, then attaches the shadow to the ball, and the depth follows from where the shadow meets the floor. No single cue carries that conclusion: it comes from a model in which one light source, one floor and one object explain the whole image. The movies are at <https://www.youtube.com/watch?v=XZ08-Df3mxA>.

A cast shadow decides the perceived motion

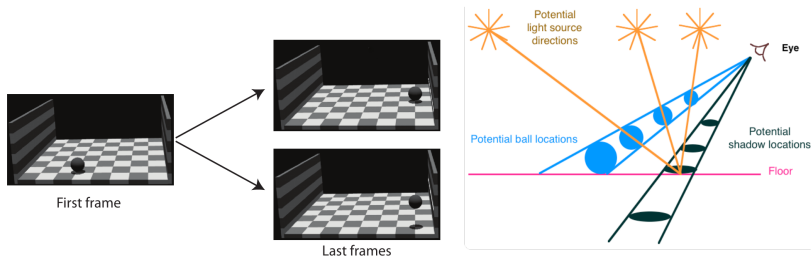


Figure 18: The first frame and the last frames of the two movies, and the family of ball positions and shadow positions consistent with one light source.

Two causes and one effect

Your friend says he is psychic and predicts that a coin toss will come up heads, and it does. Either he is psychic or the coin is two-headed, and a two-headed coin is far more common.

The graph has two causes and one shared effect. The two causes are independent a priori, which is the missing edge between them, and observing the shared effect makes them dependent. Take $P(\text{two-headed}) = 0.20$ and $P(\text{psychic}) = 0.001$, with a fair coin otherwise. After seeing heads,

$$P(\text{two-headed} \mid \text{heads}) = 0.333, \quad P(\text{psychic} \mid \text{heads}) = 0.0017.$$

Both rise, because both explain the observation. The ordinary explanation rises to a third; the extraordinary one is still far from believable.

Two causes and one effect

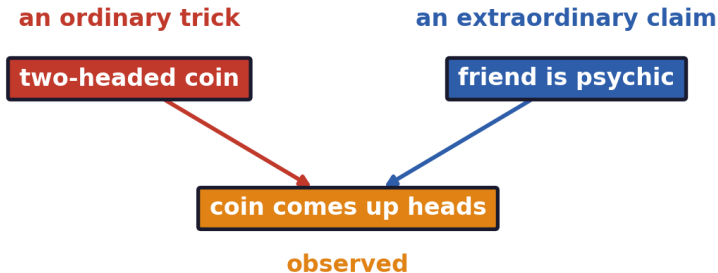


Figure 19: The two causes are independent until the shared effect is seen.

A second effect explains the first away

He is challenged to levitate a pencil, which is easy for a psychic and hard otherwise: take $P(\text{pencil} \mid \text{psychic}) = 0.99$ and $P(\text{pencil} \mid \text{not psychic}) = 0.0001$. The pencil is a second effect of the same cause, so the graph gains one edge.

	after heads	after heads and pencil
psychic	0.0017	0.943
two-headed	0.333	0.208

If the pencil levitates, the psychic hypothesis becomes probable and takes over the coin result, and the two-headed coin falls back towards its prior value of 0.20. Nothing was observed about the coin itself.

This is explaining away: evidence for one cause of an event lowers the probability of the other cause. It is a dependency created by observation, not by an edge.

A second effect explains the first away

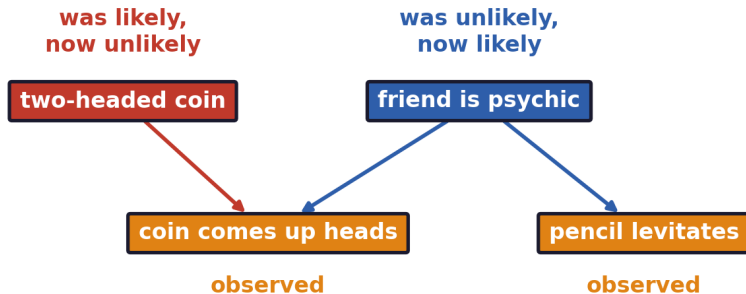


Figure 20: Once the pencil levitates, the psychic hypothesis accounts for the coin as well, and the coin hypothesis is no longer needed.

Burglary, earthquake and where you live

An alarm can be set off by a burglary or by an earthquake, and on any given night both are rare. Pearl's tables are $P(\text{burglary}) = 0.001$ and

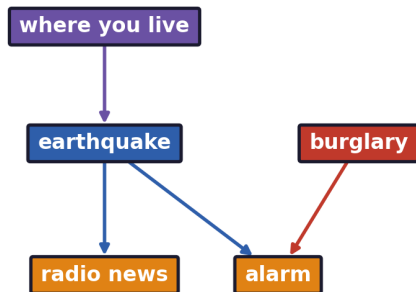
$P(\text{alarm} \mid \cdot)$	earthquake	no earthquake
burglary	0.99	0.95
no burglary	0.30	0.001

with radio news reported at $P(\text{radio} \mid \text{earthquake}) = 0.95$ and 0.001 otherwise. Where you live sets the earthquake rate, and that one number decides which of the two causes is the commoner and how much work the radio can do.

	alarm	alarm and radio news
California, $P(E) = 0.01$	0.193	0.004
Baltimore, $P(E) = 0.00005$	0.484	0.061

Radio news of an earthquake explains the alarm away, and the location sets how often that happens.

Burglary, earthquake and where you live



In California

an earthquake is possible, so radio news can take the alarm off the burglary

In Baltimore

there is no competing cause, so the alarm keeps pointing at a burglary

Figure 21: Where you live enters as a parent of the earthquake node, and it decides how much of the alarm the radio can take away.

The factorisation reduces the numbers to learn

The four variables factorise along the edges,

$$P(X_A, X_B, X_E, X_R) = P(X_A | X_B, X_E) P(X_R | X_E) P(X_B) P(X_E),$$

and any question is answered by summing out the variables of no interest,

$$P(X_B | X_A, X_R) = \sum_{X_E} P(X_B, X_E | X_A, X_R).$$

Counting the free parameters: the alarm needs four numbers, one per setting of its two parents; the radio needs two; the burglary and the earthquake need one each. Eight numbers in all. One general table over four binary variables would need $2^4 - 1 = 15$.

The saving is small here and large in general. A general table over N binary variables needs $2^N - 1$ numbers, while a graph in which no node has more than k parents needs at most $N2^k$.

The factorisation reduces the numbers to learn

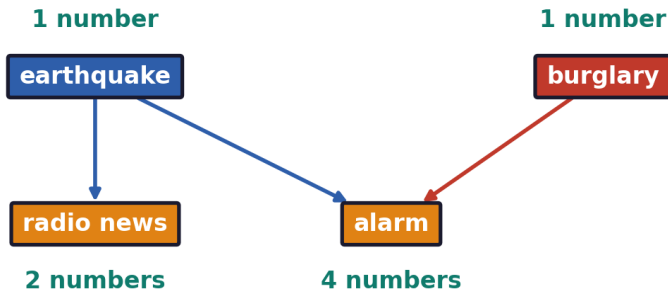


Figure 22: One small table for each node, sized by the number of parents that node has.

Explaining away in vision

With part of the T missing, observers see two separate strokes. A grey smudge placed over the gap explains why part of the T is hidden, and the T is seen. The smudge is a second cause of the same image evidence, and it takes the evidence away from the two-strokes explanation.

The Kanizsa triangle works the same way: a white triangle lying over three black discs explains all three wedges at once, so the triangle is seen and its edges are seen where no edge was drawn.

Explaining away in vision



Figure 23: The smudge explains why part of the T is hidden, so the T is seen. The Kanizsa triangle explains the three wedges.

Competing explanations of one image

A Lambertian sphere is viewed binocularly, and the highlight is placed at three different depths. At the centre of the sphere it is seen as a transparent bulb; between the centre and the surface it is seen as a shiny sphere; in front of the sphere it is seen as a cloud in front of a matt sphere.

Small changes switch the percept between models, as they do for two rotating transparent planes, where a change in the dot motion switches the percept between one rigid object and two surfaces. What changes is the model that best explains the image, and the estimate of shape follows the model that wins.

Competing explanations of one image

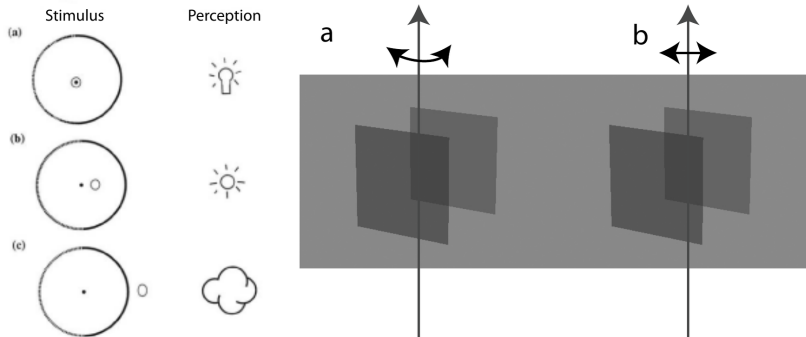


Figure 24: The stimulus, the percept it produces, and the corresponding case for two rotating transparent planes.

Model selection and explaining away

The occluded T has two models: two strokes plus a blob, or a whole T hidden behind the blob. The first needs the blob to land exactly in the gap, which would be an accident of viewpoint: move the observer slightly and the coincidence is destroyed. The second explains the image for a whole family of blob positions.

Integrating over the nuisance variable is what makes the difference.

Model 1 puts its probability on a narrow set of blob positions, so its evidence $P(I | M_1) = \int P(I | \text{blob}, M_1)P(\text{blob} | M_1)$ is small. Model 2 spreads over a wide set and collects more. The Kanizsa triangle follows the same argument.

Model selection and explaining away

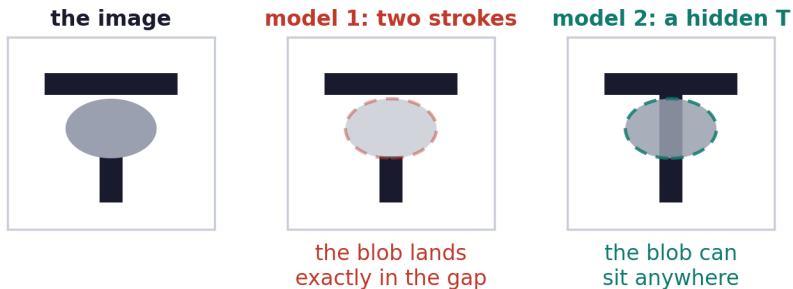


Figure 25: Model 1 needs an accident of alignment, so model 2 carries almost all the posterior.

The flying carpet revisited

The percept needs the ground plane, the segmentation of the woman and towel, and the shadow. The generic viewpoint assumption says a shadow of that shape in that place would be an accident unless something cast it. The object that cast it lies outside the image, so the shadow is attached to the towel and the towel is seen to hover.

Freeman's formulation makes the assumption quantitative: prefer the model whose explanation survives a small change of viewpoint. A model that requires a particular alignment is penalised in proportion to how narrow that alignment is.

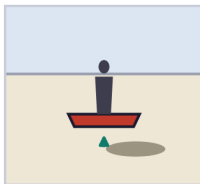
The flying carpet revisited

what reaches the eye



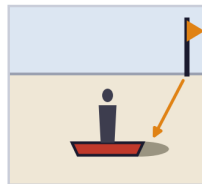
no visible caster

what is seen



the towel hovers

what is really there



a flag outside
the frame

Figure 26: What reaches the eye, what is seen, and what is really there.

Each cue is valid only in part of the image

Shape from shading works where one source lights a smooth convex surface. It fails where two surfaces reflect light onto each other, because the brightness at a point then depends on the shape elsewhere. Shape from texture works for a regular texture and fails for an irregular one, because the cue is the departure of the measured element statistics from the assumed uniform ones.

Cue combination therefore has to decide first where in the image each cue can be trusted, and that decision is itself a choice between models.

Each cue is valid only in part of the image

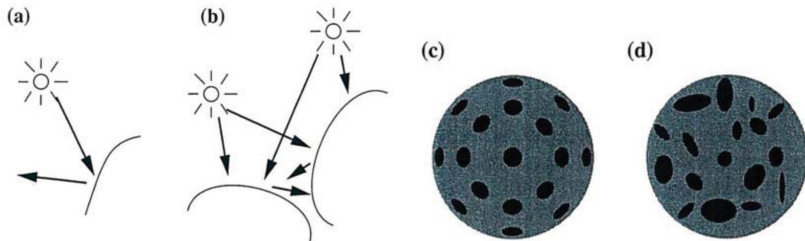


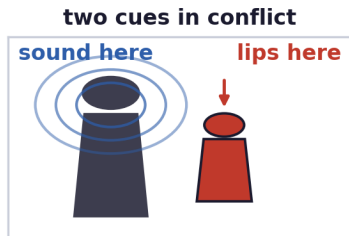
Figure 27: Two lighting configurations and two textures. Each cue is reliable in the first case of its pair and unreliable in the second.

The ventriloquism effect - audio and vision

Sight and sound sometimes have one cause, such as a barking dog, and sometimes two, such as a voice from a radio while a face moves elsewhere. A ventriloquist makes the audience treat two causes as one.

Once the two signals are attributed to one cause they are combined by the rule of Section 1, and the voice is heard at the puppet, because vision is the more precise cue for position. The decision that comes first is whether to combine the two signals at all.

The ventriloquism effect - audio and vision



vision is the more precise cue

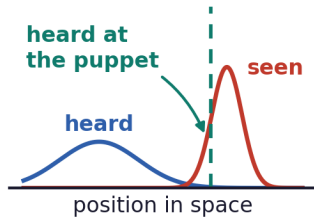


Figure 28: Two cues in conflict, and the combination that follows once they are attributed to one cause.

A meta-variable for the number of causes

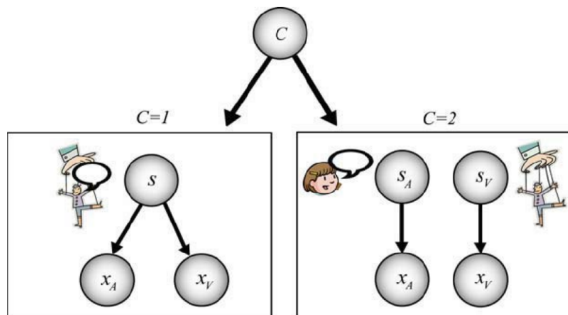
Add a variable C above the scene variables that says how many causes there are. With x_A the auditory measurement and x_V the visual one,

$$P(x_A, x_V | C=1) = \int dS P(x_A | S)P(x_V | S)P(S),$$

$$P(x_A, x_V | C=2) = \iint dS_A dS_V P(x_A | S_A)P(x_V | S_V)P(S_A)P(S_V).$$

Under $C=1$ one scene variable has to account for both measurements, so the two must agree; under $C=2$ they are free to disagree. The graph is no longer fixed: C selects which of two graphs generated the data.

A meta-variable for the number of causes



Do the sound and the sight have one cause, or two?

$C=1$ one common cause, as when a dog barks

$C=2$ two separate causes, as for a ventriloquist

Figure 29: The meta-variable and the two graphs it selects between.

The causal-inference model in closed form

With Gaussian likelihoods of variances σ_A^2 and σ_V^2 and a Gaussian prior of variance σ_p^2 centred on straight ahead, both integrals are Gaussian. Write t_A, t_V, t_p for the precisions and t_Σ for their sum. Then

$$P(x_A, x_V \mid C=1) \propto \frac{1}{\sqrt{t_\Sigma}} \exp \left\{ -\frac{1}{2} \frac{t_A t_V}{t_\Sigma} (x_A - x_V)^2 + \dots \right\},$$

so the evidence for one cause falls off as the two measurements separate, while the evidence for two causes carries no term coupling x_A to x_V : it factorises into one integral per modality. The posterior over C follows from Bayes' rule with a prior $P(C=1) = p_{\text{common}}$. The $\propto$ may hide only factors common to both models: anything differing between $C=1$ and $C=2$, the prefactor included, must be carried.

Fitting the four numbers to one subject gives values of the order

$$\sigma_V = 2.1^\circ, \quad \sigma_A = 9.2^\circ, \quad \sigma_p = 12.3^\circ, \quad p_{\text{common}} = 0.28.$$

Vision localises about four times more finely than hearing, a factor of about 19 in precision since $t = 1/\sigma^2$; one cause is the minority hypothesis before any measurement arrives.

Nearby cues are judged to have one cause

Subjects report one cause most often when the two stimuli are close together, and the report falls away as the separation grows.

On trials judged to have one cause, vision captures the heard position almost completely, and the capture is about the same at every separation. On trials judged to have two causes the bias turns negative, and the sound is heard pushed away from the light. The causal inference model follows both curves with the four fitted numbers σ_A , σ_V , σ_p and p_{common} .

Nearby cues are judged to have one cause

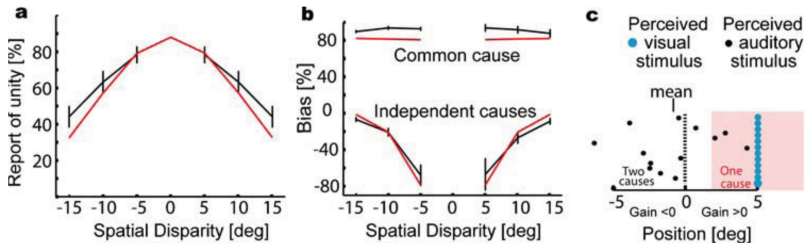


Figure 30: The report of unity against spatial disparity, the bias split by the reported number of causes, and the trials sorted by the perceived position.

Model selection versus model averaging

There are two ways to use the posterior over C . Model selection commits to the most probable model and estimates with that one,

$$C^* = \arg \max_C P(C \mid x_A, x_V), \quad P(S_A, S_V) \approx P(S_A, S_V \mid x_A, x_V, C^*).$$

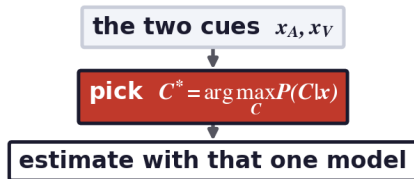
Averaging keeps both models alive and weights them by their posterior probabilities,

$$P(S_A, S_V \mid x_A, x_V) = \sum_C P(S_A, S_V \mid x_A, x_V, C) P(C \mid x_A, x_V).$$

Averaging is the correct marginal, and selection is an approximation to it that is exact when one model has nearly all the posterior. The two differ most at intermediate separations, where neither model dominates, and that is where a measurement can tell them apart.

Model selection versus model averaging

model selection



model averaging

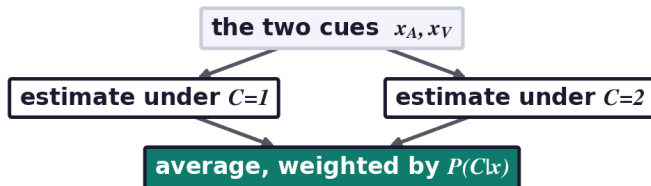


Figure 31: Committing to one model, against keeping both and weighting them by their posterior.

The chest clinic network

Eight binary variables: two risk factors, three diseases, one combined node and two findings. Every edge is a direct dependency, and the missing edges are the content of the model. A visit to Asia raises the tuberculosis rate; smoking raises lung cancer and bronchitis; tuberculosis and lung cancer both produce the same shadow on an X-ray, which is why they share one node; that node and bronchitis both produce shortness of breath.

Lauritzen and Spiegelhalter introduced this network as the worked example of their algorithm for computing with such graphs.

The chest clinic network

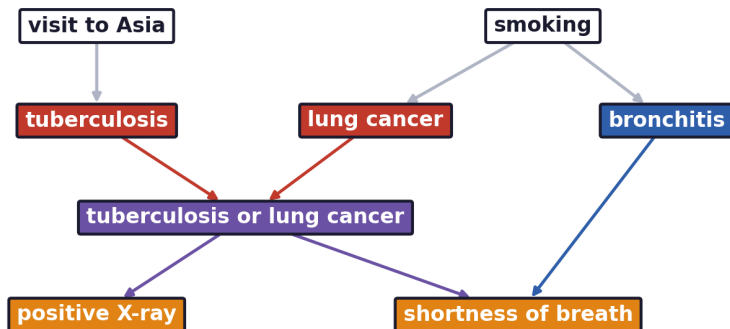


Figure 32: Eight variables and eight edges. The absent edges are assertions: smoking does not act on tuberculosis, and a visit to Asia does not act on the lung.

One small table for each node

The joint distribution factorises along the edges,

$$P(\alpha, \sigma, \tau, \lambda, \beta, \varepsilon, \xi, \delta) = P(\alpha) P(\sigma) P(\tau | \alpha) P(\lambda | \sigma) P(\beta | \sigma) \\ \times P(\varepsilon | \tau, \lambda) P(\xi | \varepsilon) P(\delta | \varepsilon, \beta),$$

with α the visit to Asia, σ smoking, τ tuberculosis, λ lung cancer, β bronchitis, ε the combined node, ξ the X-ray and δ the shortness of breath. Eighteen conditional probabilities specify the whole network, in place of $2^8 - 1 = 255$ for a general table. Only fourteen are free: ε is the logical or of τ and λ , so its four entries are fixed at 0 and 1.

One small table for each node

variable	its parents	numbers
visit to Asia	none	1
smoking	none	1
tuberculosis	visit to Asia	2
lung cancer	smoking	2
bronchitis	smoking	2
tuberculosis or lung cancer	tuberculosis, lung cancer (a fixed OR)	4
positive X-ray	tuberculosis or lung cancer	2
shortness of breath	tub. or lung cancer, bronchitis	4
the whole network	against 255 for one general table	18

14 free

Figure 33: Each node needs one number per setting of its parents.

Explaining away in the chest clinic

A positive X-ray raises both tuberculosis and lung cancer, and lung cancer is the more probable, because smoking is common and a visit to Asia is not. A recent visit to Asia then raises tuberculosis, which explains the X-ray on its own, and lung cancer falls:

	tuberculosis	lung cancer
a positive X-ray	0.092	0.489
a positive X-ray and a visit to Asia	0.338	0.371

Lung cancer falls although nothing was observed about smoking or about the lung. The occluded T, the burglary alarm and this network share the same three-node shape: two causes with one observed effect between them.

Explaining away in the chest clinic

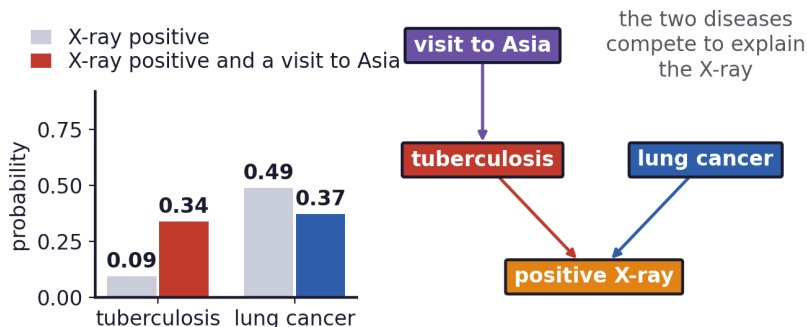


Figure 34: A positive X-ray, and the same X-ray with a visit to Asia. The two diseases compete to explain the finding.

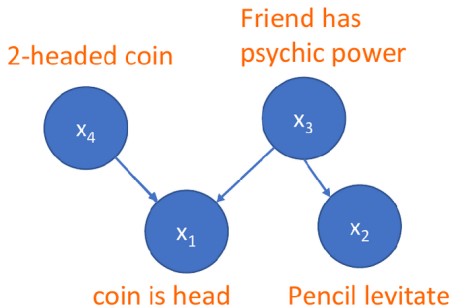
The structure makes inference cheap

Answering a question means summing out the variables the question does not ask about. On the four-node coin graph, computing $P(x_1=1)$ by summing the joint directly costs $2^3 = 8$ terms. Factorising first and moving each sum inwards leaves

$$\begin{aligned} P(x_1=1) &= \sum_{x_2} \sum_{x_3} \sum_{x_4} P(x_1=1 \mid x_3, x_4) P(x_2 \mid x_3) P(x_3) P(x_4) \\ &= \sum_{x_3} \sum_{x_4} P(x_1=1 \mid x_3, x_4) P(x_3) P(x_4) \underbrace{\sum_{x_2} P(x_2 \mid x_3)}_{\text{this sum is one}}, \end{aligned}$$

which is $2^2 = 4$ terms. What the saving costs is set by how many variables end up inside one sum. On a chain of N nodes with k states each, every sum involves two of them and the total is about Nk^2 operations, against k^{N-1} terms for the direct sum. On a general graph the cost is k raised to the size of the largest group that has to be summed together, and the next frame says what fixes that size.

The structure makes inference cheap



summing the four variables at once

$2^3 = 8$ **terms**

summing one variable at a time

$2^2 = 4$ **terms**

a chain of N nodes, k states each

k^{N-1} against about Nk^2

Figure 35: A variable that no question depends on is summed out once, and its sum is one.

Graphs with loops are grouped into a tree

The saving above needs a graph with no loops once the arrows are rubbed out. Being directed is not enough: a directed graph has no directed cycles by construction and can still have plenty of undirected ones.

The chest clinic has one. Smoking feeds lung cancer, lung cancer feeds the combined node, the combined node feeds shortness of breath, bronchitis feeds shortness of breath, and smoking feeds bronchitis, which closes the loop $\sigma - \lambda - \varepsilon - \delta - \beta - \sigma$. Tuberculosis and lung cancer are not the offenders: they have no common ancestor, one descending from the visit to Asia and the other from smoking.

Graphs with loops are grouped into a tree

The remedy is to group the nodes into clusters that themselves form a tree and to run the same recursion on the clusters. Three steps make the clusters: join the parents of every node to each other and drop the arrows; add further edges until no loop of four or more nodes is left without a chord; then take the maximal cliques of the result.

The diamond $x_1 \rightarrow x_2, x_1 \rightarrow x_3, x_2 \rightarrow x_4, x_3 \rightarrow x_4$ gains the edge $x_2 - x_3$ at the first step, because x_2 and x_3 are both parents of x_4 , and its cliques are the two clusters $\{x_1, x_2, x_3\}$ and $\{x_2, x_3, x_4\}$, joined by the nodes they share.

The cost is k raised to the size of the largest cluster, so for many graphs the tree is correct but too large to compute with. This local computation is the content of the Lauritzen and Spiegelhalter paper, and the chest clinic is its worked example.

Summary

- ▶ A directed graph names the variables and says which of them directly influence which. The joint distribution is the product of one conditional distribution per node, given its parents.
- ▶ Two conditionally independent Gaussian cues with a uniform prior combine by inverse-variance weights, and the precisions add.
- ▶ Cues that share a prior are combined through the joint posterior, so the prior is applied once. Averaging the two single-cue estimates applies it twice and keeps the estimate too close to the prior.
- ▶ When cues depend on each other, or when the model itself is in question, the coupling is strong and the estimate has to come from the joint model.
- ▶ Observing a shared effect makes independent causes dependent, which is explaining away.
- ▶ A meta-variable over models turns the choice between explanations into an inference, and the answer can be selected or averaged.
- ▶ Inference uses the same structure: linear in the number of nodes on a tree, and exponential in the size of the largest cluster in general.
- ▶ Undirected graphs replace the parent conditionals by potentials on the cliques, on the edges in the pairwise case, which is the subject of the next set of notes.

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